

Ata Utku Özkan

PhD Candidate, Computational Physics & Materials Science

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Professional Summary

- PhD computational physicist (Bilkent UNAM, exp. 2026) with 6+ years of all-atom and coarse-grained MD in LAMMPS and GROMACS; first author of a JPCB 2023 paper on [5]rotaxane conformational kinetics and a manuscript under review in *Physical Review Materials* on non-equilibrium ionic transport in 2D van der Waals solids; second author of a bioRxiv 2026 preprint on coarse-grained nuclear lamina self-assembly. Core methods: NEMD, free-energy perturbation, force-field parametrisation, enhanced sampling, statistical mechanics, SLURM/HPC.
- Production-grade Python and C++ developer: numerical analysis (NumPy, SciPy, pandas), ML (PyTorch, scikit-learn), graph analysis (NetworkX, KDTree), cheminformatics (RDKit), HPC automation (SLURM); open-source CADDack repository covering QSAR, molecular docking, and GNN-based binding affinity prediction (github.com/Ata-gith).

Technical Skills

Programming & Data	Python (NumPy, pandas, scikit-learn, PyTorch, Tensorflow/TF-Probability), C/C++ (multithreading, MPI), Bash, R, SQL (MySQL, SQLite), $\LaTeX$, Git, Linux/HPC (SLURM)
MD Simulation & Free Energy	LAMMPS, OpenMM, GROMACS, NAMD; enhanced sampling (replica exchange, umbrella sampling, metadynamics); alchemical free-energy methods (FEP, thermodynamic integration); coarse-grained and all-atom force fields (AMBER, CHARMM, MARTINI)
Cheminformatics & Structure-Based Design	RDKit, Open Babel, MDAAnalysis; molecular docking (AutoDock Vina, GNINA); QSAR modeling (scaffold-split CV, applicability domain); ADMET prediction (DeepChem); interaction fingerprints (ProLIF)
Machine Learning for Chemistry	Graph neural networks (PyTorch Geometric, DGL-LifeSci, AttentiveFP/MPNN); ML on MD trajectories; QSAR/QSPR pipelines; stochastic modeling and statistical mechanics
Domain: Materials & Quantum Chemistry	DFT (VASP, Quantum Espresso, ORCA); continuum mechanics simulation; finite-element methods (NGSolve, OpenFOAM); electronic structure and ionic transport in 2D materials
Languages	English (advanced), Turkish (native); German and French (A2)

Research & Professional Experience

2022 – present **PhD Researcher** — Bilkent University, UNAM, Ankara

Supervisor: Prof. Aykut Erbaş – Dept. Materials Science and Nanotechnology

▷ **Ionic transport in 2D van der Waals solids (first author, under review *Phys. Rev. Mater.*)** Ran all-atom non-equilibrium MD simulations (100+ runs, up to 100 ns each) of Na⁺-intercalated layered MnO₂ in LAMMPS on TRUBA HPC to resolve the microscopic origin of anomalous conductivity observed experimentally. Demonstrated that applied fields drive spatial segregation of water into hydrated and dehydrated ionic domains, producing non-monotonic conductivity and a collective ionic conduction regime that violates the Nernst–Einstein relation. Decoupled morphology from hydration effects via frozen-layer control simulations.

▷ **Nuclear lamina self-assembly (second author, *bioRxiv* 2026)** Built the Python analysis pipeline for a coarse-grained MD study of lamin protein self-assembly (1250-chain system, 22500-bead elastic nuclear shell, LAMMPS): KDTree-based network reconstruction, face-size extraction, SASA coverage metrics, NetworkX edge/face statistics. Pipeline reproduced 3D-SIM experimental lamina topology data and enabled construction of a 2D interaction-strength phase diagram.

▷ All simulation batches dispatched via SLURM; trajectory analysis automated end-to-end in Python (NumPy, pandas, SciPy, NetworkX, OVITO).

2019 – 2022 **MS Researcher** — Bilkent University, UNAM, Ankara

Thesis: Molecular Dynamics Simulations Study of Rotaxane in Bulk and at Interfaces

▷ **[5]Rotaxane conformational switching kinetics (first author, *JPCB* 2023)** Used all-atom MD (GROMACS) across seven protonation states to show that pH-driven charge changes produce sub-10 ns conformational switching of the rotaxane via CB6 ring shuttling along molecular axes. Extracted switching timescales and free-energy barriers ($U = 1\text{--}3$ kBT) from exponential fits to SASA time traces and Kramers theory; validated results across 20 replicas and two independent force fields.

▷ Simulated polyrotaxane 2D network formation at a water–chloroform interface; characterised network porosity and demonstrated charge-state-dependent lateral expansion consistent with experiment.

Mar – Jun 2017 **Computational Risk Analysis Intern** — Nurol Makine ve Sanayi A.Ş.

Statistical forecasting and computational risk modelling, quality control department.

Jan – Mar 2017 **Research Intern** — METU Nanomagnetism Instruments

Atomic force microscopy imaging and quantitative surface analysis.

Jan – Mar 2016 **Engineering Intern** — Aydokum A.Ş.

Continuum mechanics simulation and microstructure phase data analysis.

Education

2022 – present

■ **PhD, Materials Science and Nanotechnology** — Bilkent University

National Nanotechnology Research Center (UNAM), Ankara

Dissertation: Molecular Dynamics Study of Ionic Transport and Self-Assembly in Soft and 2D Materials

2019 – 2022

■ **M.Sc., Materials Science and Nanotechnology** — Bilkent University

National Nanotechnology Research Center (UNAM), Ankara

Thesis: Molecular Dynamics Simulations Study of Rotaxane in Bulk and at Interfaces

2014 – 2019

■ **B.S., Materials Science and Nanotechnology Engineering** — TOBB ETU, Ankara

Thesis: Computational Modeling of Peptide–Graphene Interactions

Fellowships & Scholarships

- 2025  **TÜBİTAK 2211-C** – National PhD Scholarship in Priority Fields in Science and Technology
- 2024  **TÜBİTAK 2224-A** – Grant for Participation in Scientific Meetings Abroad
-  **TÜBİTAK 2250** – Performance-Based Scholarship for PhD Scholars

Talks & Certifications

Invited and Contributed Talks

- 2024  **CCP5 Annual General Meeting 2024**
Molecular Dynamics Simulations of Ionic Conductivity in 2D Birnessite
agm44.ccp5.ac.uk
-  **14th Chemical Physics Congress**
Molecular Dynamics Simulation of Ionic Conductivity in 2D-MnO Systems
<https://cpc14.trakya.edu.tr>

Certifications

- 2022  **Machine Learning** — Stanford Online / Coursera
<https://www.coursera.org/account/accomplishments/verify/T65HY2KGSJLZ>
-  **Multithreading with Modern C++; ML and Data Science Bootcamp; Advanced Modern C++** — Udemy
- 2021  **Object-Oriented Data Structures in C++** — University of Illinois Urbana-Champaign / Coursera
<https://www.coursera.org/account/accomplishments/verify/HR9HNA5EP5UK>

Publications

Peer-Reviewed Journal Articles

- 1 **A. U. Özkan**, D. Tuncel, and A. Erbaş, “Effect of charge state on the equilibrium and kinetic properties of mechanically interlocked [5]Rotaxane: A molecular dynamics study,” *The Journal of Physical Chemistry B*, vol. 127, no. 5, pp. 1254–1263, 2023, PMID: 36716388.  DOI: 10.1021/acs.jpcc.2c07645.

Manuscripts Under Review

- 1 **A. U. Özkan**, T. S. Kasırga, and A. Erbaş, “Ionic transport in spontaneously ion-intercalated van der Waals layered structures,” 2026, Manuscript under review. arXiv: 2601.14836 [cond-mat.mtrl-sci].
 URL: <https://arxiv.org/abs/2601.14836>.

Preprints

- 1 H. A. Hameed, **A. U. Ozkan**, and A. Erbas, “Inter-laminar interactions control meshwork topology in a polymer-gel model of nuclear lamina,” *bioRxiv*, 2026, Preprint.  DOI: 10.64898/2026.03.14.711786.

References

Available on Request